ADARSH S

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Professional Summary

Motivated and disciplined MCA graduate with a strong foundation in programming, data structures, computer vision, machine learning, and full-stack engineering, supported by hands-on internship experience in deciphering historical manuscripts, cloud systems, and DevOps practices. Well suited to structured corporate environments, cross-functional collaboration, and the delivery of reliable, secure, and scalable solutions aligned with organizational goals. Committed to continuous learning and to contributing effectively to challenging IT projects.

Technical Skills

Programming Languages & Web: Python, Java, C, JavaScript (ES6+), PHP, SQL, MATLAB, HTML5, CSS3, JSON, HTML5 Canvas API, WebGPU, Web Audio API

AI, Computer Vision & Machine Learning: Computer Vision (OpenCV, Dlib, 3D Photometric Stereo, Sauvola Binarization), Deep Learning (PyTorch, TensorFlow, Keras, CNN, TrOCR, ResNet-34), Machine Learning (Scikit-learn, SVM, Random Forest, GNB, k-NN, Betti Invariants), Optical Character Recognition (OCR), NLP (Levenshtein Distance, Trie Lexicon, Sandhi Grammar), Biometric Verification, Image Segmentation, NumPy, SciPy, Pandas, Matplotlib, Seaborn

Frameworks & App Development: Django, Flask, Node.js, React.js, Laravel, Flutter, Android Development, RESTful APIs, ReportLab (PDF Engine), KaTeX, Responsive Web Design, Front-End & Full-Stack Development

Databases, Cloud, DevOps & Tools: MySQL, MongoDB, Database Development, Cloud Computing, DevOps (Docker, Git, GitHub, CI/CD), Linux/Unix, Anaconda, PyCharm, MySQL Workbench, Postman, Version Control

Emerging Tech & Security: Blockchain, Smart Contracts, Internet of Things (IoT), Industrial IoT (IIoT), Cybersecurity, Computer Networking, Operating Systems, Privacy-by-Default (GDPR), Substrate Verification

Core Competencies & Professional Skills: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Document Intelligence, Palaeographic Chronometry, Epigraphical Restoration, Problem Solving, Project Planning, Cross-Functional Team Collaboration

Professional Experience

Bluegen Solutions | Software Development Intern — Dec 2025 – Mar 2026
Kozhikode, Kerala, India (On-site)

- Engineered scalable backend application modules using Python and Django, implementing RESTful API endpoints and secure authentication to streamline server-side business logic.
- Designed and optimized a normalized MySQL relational schema across 8+ tables with indexing and foreign-key constraints, reducing query retrieval latency by 25%.
- Collaborated within an Agile engineering team to profile, debug, and resolve 15+ critical software issues, boosting overall application reliability and stability.

The Skybrisk | React.js Development Intern — Feb 2026 – Mar 2026
Remote

- Developed core enterprise ERP web modules (Product Catalog, Purchase/Sales Orders, GRN, and Invoicing) utilizing React.js, Redux Toolkit, and Material-UI for responsive business operations.
- Constructed 15+ reusable modular UI components and integrated RESTful APIs with token-based JWT route guards (RBAC), slashing front-end code duplication by 35%.
- Implemented interactive data visualization dashboards using Chart.js and client-side PDF invoice export via jsPDF, enhancing reporting speed and operational efficiency by 20%.

CODTECH IT Solutions | DevOps Intern — Feb 2026 – Mar 2026
Remote

- Automated end-to-end CI/CD build and test workflows using GitHub Actions, streamlining release cycles and establishing standardized version control with zero-downtime deployment.
- Orchestrated containerized microservices deployments with Kubernetes (K8s) manifests, configuring multi-replica pods (3 replicas) and Service LoadBalancers for high availability.
- Integrated DevSecOps pipelines incorporating automated OWASP ZAP vulnerability baseline scans (DAST), identifying security misconfigurations and enforcing automated compliance checks.

Academic Projects

AssentTag — Dec 2025 – Mar 2026

Technologies: Python, Django, OpenCV, Dlib, ResNet-34, MySQL, NumPy, JavaScript (ES6+), HTML5/CSS3, REST APIs, Git

- Engineered an automated privacy-by-default vision pipeline using OpenCV and Dlib to execute real-time facial detection and apply selective Gaussian blurring to safeguard unconsented individuals in shared media.
- Implemented biometric identity verification leveraging ResNet 128-dimensional facial embeddings and Euclidean distance metric thresholding to validate live camera feeds and prevent fraudulent accounts.
- Architected a high-throughput Django and MySQL backend with thread-safe processing and in-memory descriptor caching, orchestrating dynamic consent workflows ("The Veil" unblurring) and live notifications.
- Developed a responsive full-stack web interface featuring ephemeral stories, granular user privacy controls, multi-factor OTP authentication, and automated GDPR-compliant audit report generation.

EpigraphiX-AI: Neural Palm-Leaf Manuscript OCR & Epigraphical Intelligence Suite Jul 2025 – Nov 2025
Technologies: Python, MATLAB, JavaScript (ES6+), Node.js, OpenCV, PyTorch, TensorFlow, Scikit-learn, SciPy, NumPy, CNN, SVM, Random Forest, k-NN, GNB, NLP, ReportLab, HTML5 Canvas, Git, TrOCR, WebGPU, Web Audio, KaTeX, Matplotlib

- Engineered a background-invariant vision pipeline to authenticate historical palm leaves. Integrated fast Integral Sauvola binarization and 3D stereo to reconstruct incised grooves.
- Architected a hybrid TrOCR Vision Transformer and CNN Lattice for sequence transcription. Benchmarked 5 classifiers on Betti invariants, achieving 99.5% accuracy and 97.8% precision.
- Constructed an epigraphical NLP engine with DP Levenshtein alignment and a 354-word trie. Integrated dynasty dating and 4-way translation into Modern Malayalam, English, and Hindi.
- Developed a Document Intelligence Web Studio with Web Audio chant resonance synthesis. Built an executive vector PDF export engine generating comprehensive 6-gauge scorecard analytics.

Publications

EpigraphiX-AI: Neural Epigraphical OCR, Topological Binarization, Strict Manuscript Authentication, and Multi-Model Intelligence for Degraded Historical Palm-Leaf Manuscripts August 2026
Author: Adarsh S | **Journal:** IJRASET | **Paper ID:** IJRASET84707 | **ISSN:** 2321-9653
Volume & Issue: Vol. 14, Issue VIII | **Impact Factor:** 7.429 | **DOI:** 10.22214/ijraset.2026.84707

- Formulated an end-to-end Computer Vision and Deep Learning framework incorporating Vision Transformers (TrOCR) and $O(1)$ fast Integral Sauvola binarization to transcribe incised, inkless historical palm-leaf characters.
- Engineered an organic substrate authentication module using 3D micro-depth relief profiling ($\nabla \cdot \vec{N}$) and longitudinal cellulose fiber energy to verify physical manuscripts and reject synthetic artifacts.

Education

Master of Computer Applications August 2024 – May 2026
Government Engineering College, Thrissur **CGPA: 7.33/10.0**

Bachelor of Science in Computer Science July 2019 – March 2022
College of Applied Science IHRD, Thamarassery **CGPA: 6.61/10.0**

Higher Secondary Education (Biology Stream) June 2017 – March 2019
Government Higher Secondary School, Thamarassery **Percentage: 79.9%**

High School Education June 2016 – March 2017
MGM Higher Secondary School, Engapuzha **Percentage: 98.8%**

Certifications & Achievements

- **CloudOps Engineer – Associate** – Amazon Web Services (AWS) (August 2026)
- **TCS iON Career Edge – IT Primer** – TCS iON (February 2026)
- **Introduction to Modern AI** – Cisco Networking Academy (February 2026)
- **SOAR – AI to be Aware** – Skill India Digital Hub (January 2026)
- **Technology Job Simulation** – Deloitte (January 2026)
- **Full Stack Development with React and Drupal** – Fireflies (January 2022)
- **Microsoft Excel Advanced Analytics** – Simplilearn (December 2025)
- **Deep Learning with Keras** – Simplilearn (December 2025)
- **Google Gemini Certification** – Simplilearn (December 2025)
- **Programming in Java** – NPTEL, IIT (November 2025)

- **Ministry of Youth Affairs & Sports (MYBharat)** – Ministry of Youth Affairs & Sports (September 2025)
- **SQL Basics** – HackerRank (September 2025)
- **Data Analysis Using Python** – IBM (June 2025)
- **MTA: Introduction to Programming Using Python** – ScholarHat (June 2025)
- **C Programming** – ScholarHat (June 2025)
- **Data Science** – Simplilearn (June 2025)
- **Hackathons & Achievements:** Participated in the Ideas to Impact Challenge 2024 (IEDC LBSCEK), demonstrating analytical thinking, problem-solving, innovation, and teamwork skills.